Week 10: Bridges

Adult guide built from the problem files · Unpiloted.

The answers, hints and routes are in the adult guide, week-10-facilitator.pdf, whose source is guide-src/facilitator.tex. First route = the guide's core problems.

Problems and routes

Problem	K–1	2–3	4–5	route
Pick up every counter in four towns	P1			first
Every island where a walk can start	P2			first
Can you pick up every counter?	P3			first
End on the island where you started	P4			return
Trace each picture in one stroke	P5			return
One more bridge in each town	P6			return
Build towns with hexagons and craft sticks	P7			return
A bridge with two counters	P8			return
Find a walk in each town		P1	P1	first
Where walks start and end		P2	P2	first
Decide without walking		P3		first
One new bridge so that a walk can		P4		first
Fewest new bridges for a walk that comes home		P5		return
How to tell without walking		P6		return
Trace each picture, or explain why not		P7		return
Build towns to a description		P8		return
A walk that can start on only one island?		P9		return
Second counters for a walk that comes home		P10		return
Decide without walking, and where walks start and end			P3	first
Write the rule and explain it			P4	first
The bridges of Königsberg			P5	first
Exactly one or three odd islands?			P6	return
Trace each picture, or find the fewest strokes			P7	return
Lee's walk around the outside			P8	return
Why every town that fits the rule has a walk			P9	return
The truck's fewest extra counters			P10	return
Why no route uses fewer extra counters			P11	return

First route. K–1 P1–P3; Grades 2–3 P1–P4; Grades 4–5 P1–P5. **Return visits.** K–1 P4–P8; Grades 2–3 P5–P10; Grades 4–5 P6–P11.

Key, one entry per problem

These problem files hold no keys; the week's adult guide has them.